



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

Backend Development with Node.js, Express, and Mongo

Theory and Practical Answer Paper – Set - B and Set – 2

INSTRUCTOR NAME: SANTHANALAKSHMI

INSTITUTION NAME: CHRIST (Deemed to be University)

CLASS: 5BTIT

DATE OF SUBMISSION: 23-07-2026

<u>NAME</u>	<u>REGISTRATION NUMBER</u>	<u>CHRIST EMAIL ID</u>
SRI RAMANARAYANA S N	2461029	sri.ramanarayana@btech.christuniversity.in

Department of Computer Science and Engineering

School of Engineering and Technology

CHRIST (Deemed to be University)

Kumbalgodu, Bangalore - 560 074

July-2026

THEORY ANSWER PAPER

23.07.2026

L & T Edutech

Sri Pramanarayanan SN

CIA-1 : Theory Part - Set B

2461029

5BT1T

Q1. Write a program to check whether a given year is a leap year using if-else.

```
=> let year = parseInt(prompt("Enter a year: "));  
if ((year % 400 === 0) && year % 100 !== 0) || (year % 400 !== 0)  
{  
    console.log("Leap year");  
}  
else {  
    console.log("Not a Leap year");  
}
```

Q2. The loop that checks its condition before executing its body is called a Entry-Controlled loop [for, while]

Q3. Write a program to print numbers from 1 to 20, skipping multiples of 3 using continue.

```
=> for (let i = 1; i <= 20; i++) {  
    if (i % 3 === 0) {  
        continue;  
    }  
    console.log(i);  
}
```

Q4. The method used to add a new element to the beginning of an array is unshift().

Q5. Write a program to create an object 'book' with properties title, author and price, then print each property.

```
=> let book = {  
    title: "The Odyssey"  
    author: "Homer"  
    price: 450  
}
```

```
console.log("Title:", book.title);  
console.log("Author:", book.author);  
console.log("Price:", book.price);
```

PRACTICAL ANSWER PAPER

Hospital Patient System

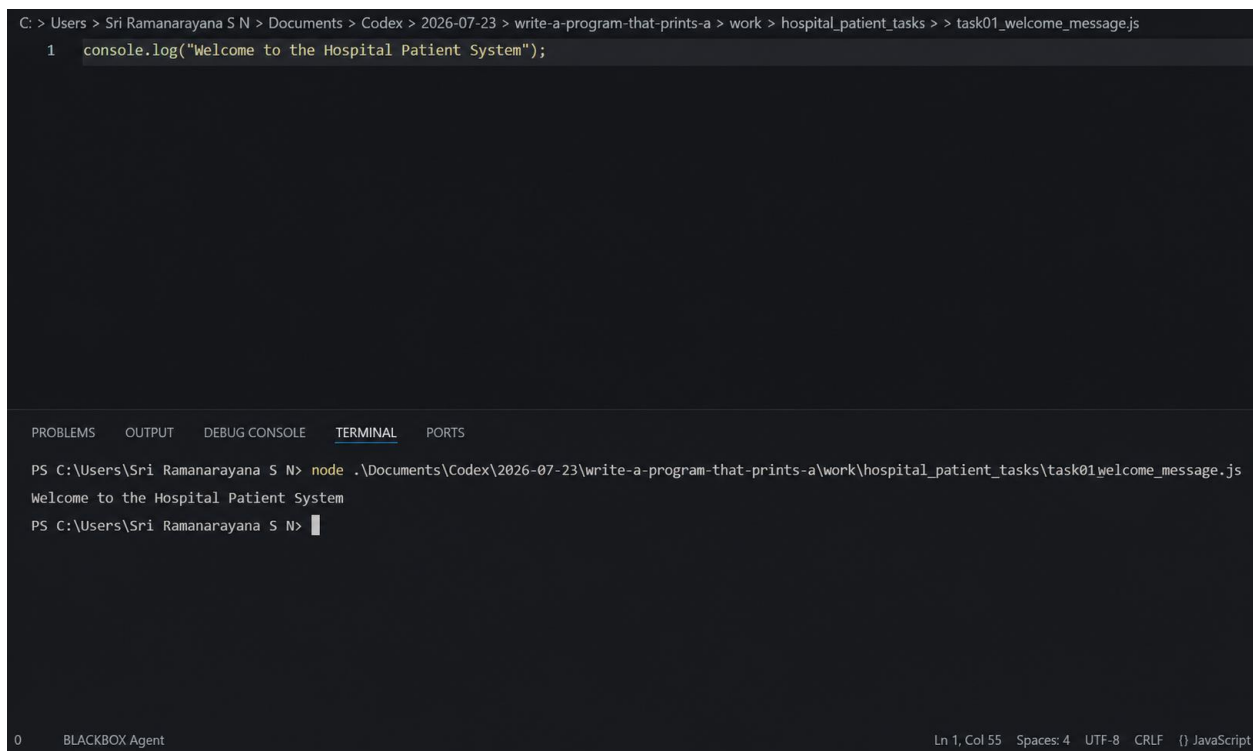
Node & Express JS CIA-1 Practical Answers

Topic	Hospital Patient System
Tasks Covered	Task 1 to Task 10
Language Used	JavaScript with Node.js

Task 1: Welcome Message

Question: Write a program that prints a welcome message for the "Hospital Patient System" to the console.

Code and Output Screenshot



```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > > task01_welcome_message.js
1 console.log("Welcome to the Hospital Patient System");
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task01_welcome_message.js
Welcome to the Hospital Patient System
PS C:\Users\Sri Ramanarayana S N> |
```

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF {} JavaScript

Output

Welcome to the Hospital Patient System

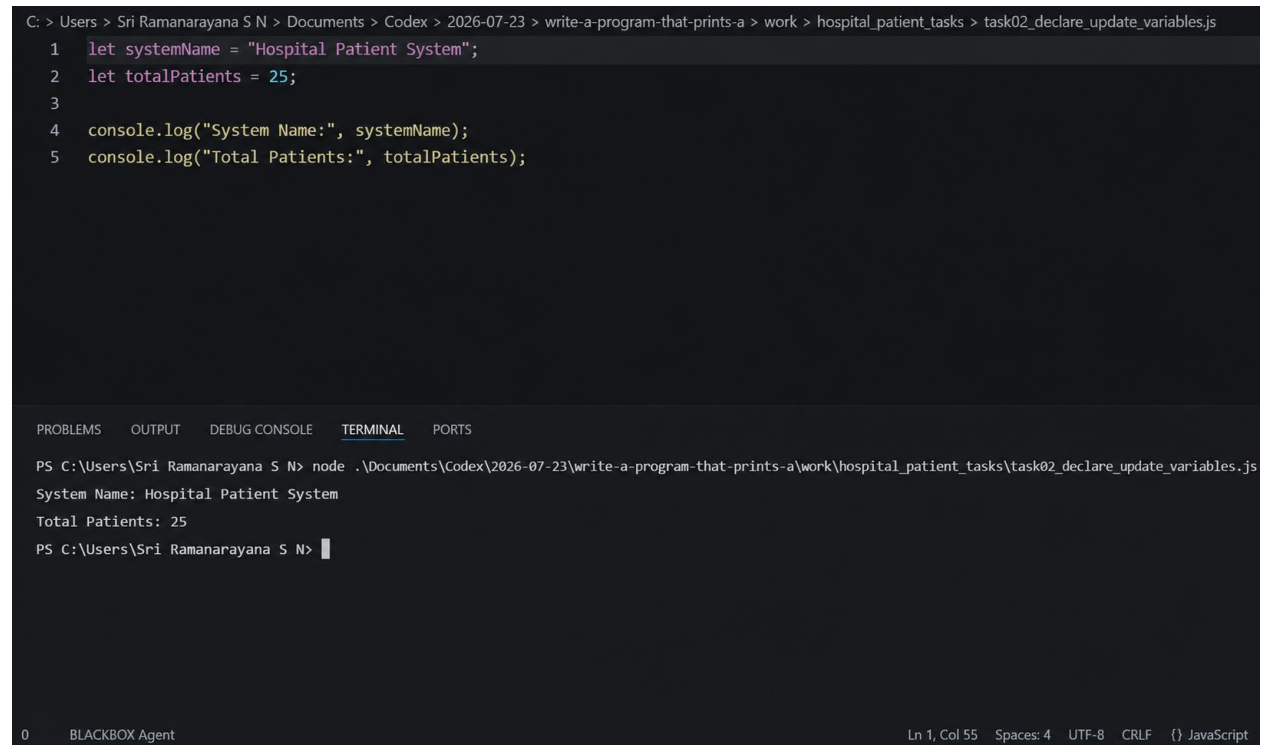
Summary: Used console.log() to display a welcome message for the Hospital Patient System.

Used: console.log()

Task 2: Declare and Update Variables

Question: Declare variables to store the system name and the total number of patients, then print both.

Code and Output Screenshot



The screenshot displays a code editor with a dark theme. The top section shows the file path: C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task02_declare_update_variables.js. The code editor contains five lines of JavaScript code: 1. let systemName = "Hospital Patient System"; 2. let totalPatients = 25; 3. (empty line) 4. console.log("System Name:", systemName); 5. console.log("Total Patients:", totalPatients);. Below the code editor is a terminal window with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the command PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task02_declare_update_variables.js and its output: System Name: Hospital Patient System and Total Patients: 25. The terminal prompt is PS C:\Users\Sri Ramanarayana S N>.

```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task02_declare_update_variables.js
1  let systemName = "Hospital Patient System";
2  let totalPatients = 25;
3
4  console.log("System Name:", systemName);
5  console.log("Total Patients:", totalPatients);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task02_declare_update_variables.js
System Name: Hospital Patient System
Total Patients: 25
PS C:\Users\Sri Ramanarayana S N> 
```

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF {} JavaScript

Output

System Name: Hospital Patient System
Total Patients: 25

Summary: Declared a string variable for the system name and a number variable for the total patients, then printed both values.

Used: let variables and console.log()

Task 3: Identify Datatypes

Question: Declare variables of type Number, String, and Boolean related to a patient, and print their datatypes using `typeof`.

Code and Output Screenshot

```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task03_identify_datatypes.js
1  let patientAge = 45;
2  let patientName = "Arun Kumar";
3  let isAdmitted = true;
4
5  console.log("patientAge datatype:", typeof patientAge);
6  console.log("patientName datatype:", typeof patientName);
7  console.log("isAdmitted datatype:", typeof isAdmitted);

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task03_identify_datatypes.js
patientAge datatype: number
patientName datatype: string
isAdmitted datatype: boolean
PS C:\Users\Sri Ramanarayana S N> █

0  BLACKBOX Agent  Ln 1, Col 55  Spaces: 4  UTF-8  CRLF  {} JavaScript
```

Output

```
patientAge datatype: number
patientName datatype: string
isAdmitted datatype: boolean
```

Summary: Created patient-related number, string, and boolean variables and used `typeof` to identify their datatypes.

Used: `typeof` operator

Task 4: Perform a Calculation

Question: Write a program to calculate the age difference between two patients using arithmetic operators, and print the result.

Code and Output Screenshot

```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task04_age_difference.js
1  let patientOneAge = 62;
2  let patientTwoAge = 37;
3
4  let ageDifference = Math.abs(patientOneAge - patientTwoAge);
5
6  console.log("Age difference between the two patients is", ageDifference, "years.");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task04_age_difference.js
Age difference between the two patients is 25 years.
PS C:\Users\Sri Ramanarayana S N> |
```

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF {} JavaScript

Output

Age difference between the two patients is 25 years.

Summary: Used subtraction and Math.abs() to calculate the positive age difference between two patients.

Used: arithmetic operator and Math.abs()

Task 5: Classify a Record

Question: Write a program to check whether a given patient's age meets a specific condition using if-else.

Code and Output Screenshot

```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > *task05_classify_record.js

1  let patientName = "Meena";
2  let age = 68;
3
4  if (age >= 60) {
5      console.log(patientName + " is eligible for senior patient care.");
6  } else {
7      console.log(patientName + " is not eligible for senior patient care.");
8  }

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task05_classify_record.js
Meena is eligible for senior patient care.
PS C:\Users\Sri Ramanarayana S N> |
```

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF {} JavaScript

Output

Meena is eligible for senior patient care.

Summary: Used an if-else condition to classify whether the patient is eligible for senior care based on age.

Used: if-else condition

Task 6: Iterate Over Records

Question: Create an array of at least 4 patient objects and use a loop to print each one's details.

Code and Output Screenshot

```
C: > Users > Sri Ramaarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task06_iterate_records.js
1  let patients = [
2      { id: 101, name: 'Arun', age: 45, department: 'Cardiology' },
3      { id: 102, name: 'Priya', age: 32, department: 'Neurology' },
4      { id: 103, name: 'Kiran', age: 58, department: 'Orthopedics' },
5      { id: 104, name: 'Sneha', age: 27, department: 'Dermatology' }
6  ];
7
8  for (let i = 0; i < patients.length; i++) {
9      console.log(
10         patients[i].id + ' - ' +
11         patients[i].name + ', Age: ' +
12         patients[i].age + ', Department: ' +
13         patients[i].department
14     );
15 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task06_iterate_records.js
101 - Arun, Age: 45, Department: Cardiology
102 - Priya, Age: 32, Department: Neurology
103 - Kiran, Age: 58, Department: Orthopedics
104 - Sneha, Age: 27, Department: Dermatology
PS C:\Users\Sri Ramanarayana S N> |
```

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF {} JavaScript

Output

```
101 - Arun, Age: 45, Department: Cardiology
102 - Priya, Age: 32, Department: Neurology
103 - Kiran, Age: 58, Department: Orthopedics
104 - Sneha, Age: 27, Department: Dermatology
```

Summary: Created an array of four patient objects and used a for loop to print each patient's details.

Used: array of objects and for loop

Task 7: Skip or Stop Early

Question: Using the array of patients, use break or continue while looping to skip or stop at a specific condition on age.

Code and Output Screenshot

```
C: > Users > Sri Ramanarayana S N> Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task07_skip_or_stop_early.js

1  let patients = [
2    { name: 'Arun', age: 45 },
3    { name: 'Priya', age: 28 },
4    { name: 'Kiran', age: 58 },
5    { name: 'Sneha', age: 24 }
6  ];
7
8  for (let patient of patients) {
9    if (patient.age < 30) {
10     continue;
11    }
12  }
13
13  console.log(patient.name + ' is ' + patient.age + ' years old.');
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task07_skip_or_stop_early.js
Arun is 45 years old.
Kiran is 58 years old.
PS C:\Users\Sri Ramanarayana S N> 
```

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF {} JavaScript

Output

Arun is 45 years old.
Kiran is 58 years old.

Summary: Used continue to skip patients below 30 years of age while printing the remaining records.

Used: for...of loop with continue

Task 8: Add and Find

Question: Use push() to add a new patient to the array. Write a loop to find the patient with the highest age.

Code and Output Screenshot

```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > J task08_add_and_find.js

1  let patients = [
2    { name: 'Arun', age: 45 },
3    { name: 'Priya', age: 32 },
4    { name: 'Kiran', age: 58 }
5  ];
6
7  patients.push({ name: 'Lakshmi', age: 72 });
8
9  let oldestPatient = patients[0];
10
11  for (let patient of patients) {
12    if (patient.age > oldestPatient.age) {
13      oldestPatient = patient;
14    }
15  }
16
17  console.log("Oldest patient:", oldestPatient.name);
18  console.log("Age:", oldestPatient.age);

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task08_add_and_find.js
Oldest patient: Lakshmi
Age: 72
PS C:\Users\Sri Ramanarayana S N> █

0  BLACKBOX Agent  Ln 1, Col 55  Spaces: 4  UTF-8  CRLF  {} JavaScript
```

Output

Oldest patient: Lakshmi
Age: 72

Summary: Used push() to add a new patient and a loop to find the patient with the highest age.

Used: push() method and comparison loop

Task 9: Remove and Sort

Question: Use `pop()` or `shift()` to remove an entry from the array. Use the array's `sort()` method to sort patients by age.

Code and Output Screenshot

```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > * task09_remove_and_sort.js
1  let patients = [
2    { name: 'Arun', age: 45 },
3    { name: 'Priya', age: 32 },
4    { name: 'Kiran', age: 58 },
5    { name: 'Sneha', age: 27 }
6  ];
7
8  let removedPatient = patients.pop();
9  patients.sort((a, b) => a.age - b.age);
10
11 console.log("Removed patient:", removedPatient.name);
12 console.log("Patients sorted by age:");
13
14 for (let patient of patients) {
15   console.log(patient.name + " - " + patient.age);
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task09_remove_and_sort.js
Removed patient: Sneha
Patients sorted by age:
Priya - 32
Arun - 45
Kiran - 58
PS C:\Users\Sri Ramanarayana S N>
```

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF () JavaScript

Output

Removed patient: Sneha
Patients sorted by age:
Priya - 32
Arun - 45
Kiran - 58

Summary: Used `pop()` to remove the last patient and `sort()` to arrange the remaining patients by age.

Used: `pop()` method and `sort()` method

Task 10: Create and Print an Object

Question: Create a patient object with at least 5 properties and print each property individually.

Code and Output Screenshot

```
C: > Users > Sri Ramanarayana S N > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task10_create_print_object.js

1  let patient = {
2      id: 201,
3      name: 'Ravi Sharma',
4      age: 40,
5      department: 'General Medicine',
6      isAdmitted: true
7  };
8
9  console.log("Patient ID:", patient.id);
10 console.log("Name:", patient.name);
11 console.log("Age:", patient.age);
12 console.log("Department:", patient.department);
13 console.log("Admitted:", patient.isAdmitted);

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Sri Ramanarayana S N> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task10_create_print_object.js
Patient ID: 201
Name: Ravi Sharma
Age: 40
Department: General Medicine
Admitted: true
PS C:\Users\Sri Ramanarayana S N> 
```

Output

Patient ID: 201
Name: Ravi Sharma
Age: 40
Department: General Medicine
Admitted: true

Summary: Created a patient object with five properties and printed each property separately using dot notation.

Used: object properties and dot notation

Git-hub Link: https://github.com/SriRam-0511/L-T_Sem5